

6th Progress Report

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Subject:	RL for Flying-V
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Outcome of the previous meeting

- Reviewed research questions (see appendix)
- Reviewed research plan, discussed the duration and contents for each work package (obtaining linear model probably doesn't take 5 working days, implementing IDHP on my own could take some time but a contingency can be to use pre-implemented IDHP algorithms from other students). One of the work package names was deemed confusing.

Description of the results since last meeting

- Finished and submitted literature review, awaiting feedback.
- Decided to prioritize developing an eligibility trace implementation instead of multi-step first. **Reasoning:** I think that this is easier to implement correctly than multi-step so this work package would probably be quicker, hopefully leaving me with more wiggle room to implement a SAC/DSAC agent or multi-step extension.
- Regarding the decision between implementing the Online-offline hybrid controller vs implementing a second augmentation to IDHP. I think good arguments can be made for both options, online-offline hybrid would provide a more robust flight controller (pure IDHP diverges sometimes esp if the control task is difficult), thus more fault tolerant and thus my work will be more true to the research questions; but improving IDHP is academically interesting, it's worthwhile to try to push the boundaries of an adaptive controller because adaptiveness holds more potential for fault tolerance and robustness, though this path is slightly less true to my research objectives in my opinion (adaptive controllers seem to have higher failure rate than DRL based controllers, more DRL controllers have been used on high fidelity simulations as well).

Frankly, I am inclined to work on pushing the abilities of adaptive controllers, but I will first see how well the eligibility trace augmentation goes. If it demonstrates significant improvement in fault tolerance or convergence rate than vanilla IDHP, then that would mean there is real potential in augmenting IDHP and therefore I will work on trying to make the multi-step IDHP work. If there is no significant improvement, then I will go with implementing the offline-online hybrid instead.

- Looked at implementations of IDHP and started to conceptualize how I can implement my own code.

- Contacted Ramzi and Direnc for models of the Flying-V, currently working out getting the model from them.
- Reworded the confusing work package name and extended the time budget by 1 to 3 days for each of the work packages in research phase 1.
- Working out Gitlab access and permissions with Direnc

Plans for next 2 weeks

1. Get and familiarize myself with the Flying-V mode from Direnc, and get the linearizing scripts from Ramzi.
2. I will probe a little on how to integrate Flight gear into my work.
3. Continue familiarizing myself with IDHP implementations, and with tensorflow functions. Probably going to make a pseudo code of the algorithm to make it communicable and for sanity checks. Keeping an eye out for how I can incorporate eligibility traces when studying how other people implemented IDHP.

A Research Questions

Research Objective

To advance the state-of-the-art reinforcement learning based flight controllers and further the technological readiness level of the Flying-V, by developing an intelligent and fault-tolerant flight control system for the Flying-V.

Research Questions

1. What reinforcement learning algorithm can yield a flight controller that is the most tolerant to faults?
 - (a) What RL algorithms are considered to be state-of-the-art?
 - (b) How is fault tolerance defined and tested in past research?
 - (c) Which algorithms have been shown to provide the best fault tolerance?
 - (d) What reference tracking performance have these algorithms shown in past research?
2. What are the flight control-related challenges when it comes to designing a controller for the Flying-V?
 - (a) What are the flight dynamic properties of the Flying-V and what implications do they have on an AFCS system?
 - (b) What are potential fault scenarios that warrant attention from the AFCS system in the Flying-V?
 - (c) What criteria and metrics should be used to characterize a controller's fault tolerance?
3. How can the identified algorithm be applied to control the Flying-V?
 - (a) How should the flight control system be structured?
 - (b) What are the variables defining the MDP in the case of controlling the Flying-V?
4. How does the implemented flight controller perform during nominal flight and in the presence of faults?
 - (a) Noting the dynamical properties and possible fault scenarios of the Flying-V, what flight scenarios should be designed to test the performance of the controller?
 - (b) What is the nominal performance and degree of fault tolerance of the implemented controller?
 - (c) How well does the nominal flight performance of the proposed flight controller compare to other state-of-the-art controllers?
 - (d) How are the fault tolerance characteristics of the proposed flight controller compared to other state-of-the-art controllers?